

Computing revealed preference goodness-of-fit measures with integer programming

Thomas Demuynck*

John Rehbeck

Ecares

Department of Economics

Université Libre de Bruxelles

The Ohio State University

Thomas.Demuynck@ulb.be

rehbeck.7@osu.edu

November 22, 2021

Abstract

This paper develops mixed-integer linear programming (MILP) formulations to compute various revealed preference goodness-of-fit measures. We provide MILP formulations to compute the Houtman-Maks Index, the Average Varian Index, and the Minimum Cost Index when there are linear budgets. Next, we provide MILPs to compute minimal “measurement error” in expenditures, prices, and quantities. Finally, we extend our results to non-linear budgets. As a proof of concept, we compute various goodness-of-fit measures for experimental choice data sets from the literature. The maximal computation time is less than 3 seconds for all measures examined on these datasets.

JEL Classification Numbers: D1

Keywords: Revealed preference; choice consistency; computation

*Thomas Demuynck acknowledges financial support by the Fonds de la Recherche Scientifique-FNRS under EOS project 30544469.

1 Introduction

This paper focuses on computing revealed preference goodness-of-fit measures for the standard consumer problem. In particular, we focus on the Houtman-Maks Index (HMI) [Houtman and Maks, 1985], the Average Varian Index (AVI) [Varian, 1990], and the Minimum Cost Index (MCI) [Dean and Martin, 2016]. We show that these measures can be formulated in terms of linear programs involving both continuous and binary variables. In addition, we introduce several new measures to capture the minimal amount of errors in expenditures, prices, or quantities to obtain rationalisability, and we show that these minimal errors can also be easily computed using mixed integer linear programs. Finally, we discuss how our framework can easily be modified to settings with nonlinear budget sets, following Forges and Minelli [2009].

Motivation Since Samuelson [1938] introduced the revealed preference methodology, it has attracted considerable attention. Subsequent research by Houthakker [1950], Richter [1966], Afriat [1967], and Varian [1982] provide conditions that easily allow one to verify whether an individual's observed choices are consistent with the maximisation of some utility function. If so, the observed choices are said to be rationalisable. It was quickly discovered that the resulting revealed preference tests are frequently refuted when applied to real life data.¹

However, revealed preference tests only provide a Yes/No answer: a dataset is either rationalisable or it is not. In most cases, such Yes/No answers are not very informative. As noted by Varian [1990], what we usually care about is whether utility maximisation is a reasonable way to describe actual behaviour. For most purposes, nearly optimizing behaviour is just as good as optimizing behaviour.

In order to assess the severity of the revealed preference violations, several researchers have proposed goodness-of-fit measures to assess how close an individual's choices are to being rationalisable. Many of these measures are difficult to compute. The contribution of this paper is to propose a unifying mixed integer programming formulation for the most popular goodness-of-fit measures. Although MILP have exponential worst time complexity, there exist several modern software packages are able to solve (moderately sizes) MILPs quite efficiently.

¹See Koo [1963] and Battalio, Kagel, Winkler, Fisher Jr, Basmann, and Krasner [1973] for early applications on household and experimental choice data.

Contribution The literature contains a wide range of different goodness-of-fit measures of choice consistency.² While some of these measures can be computed using efficient algorithms or standard linear programming methods, for others there is no efficient algorithm available. In fact, several of them are known to be computationally hard [Smeulders, Spieksma, Cherchye, and De Rock, 2014, Dean and Martin, 2016]. These computational issues pose a severe limitation for using these measures. For example, Choi, Fisman, Gale, and Kariv [2007] report that the Houtman-Maks index to be “computationally intensive” for a dataset with fifty observations. In addition, since there is no unifying approach to compute these goodness-of-fit measures, their usage is severely limited. In this paper, we fill this gap by proposing a unifying MILP approach to compute various goodness-of-fit measures.

We illustrate our results by computing the HMI, AVI and MCI for various data in the literature.³ These data sets involve individual choices ranging from 22 to no less than 79 observations per individual. Using a standard desktop computer, we find that most of the goodness-of-fit measures can be computed in less than a second while no index takes more than 2 seconds to compute. This demonstrates that the computation of these goodness-of-fit measures, although theoretically complex, can be computed quite efficiently in practice.

Next, our paper also contributes to computational methods used to address various revealed preference problems. Mixed-integer linear programs have been used to great success on a variety of revealed preference models.⁴ By formulating various goodness-of-fit measures as the solution of a MILP, our paper extends the list of applications of MILP methods in revealed preference analysis. Thereby, we strengthen the case for MILP methods as a useful part of the revealed preference toolkit. This paper is related to the research in Heufer and Hjertstrand [2015] that gives a MILP to compute the HMI. This paper extends these methods to other goodness-of-fit measures. Moreover, we extend the applicability of these methods to settings with general, non-linear budgets.

²For a non-exhaustive list, see Houtman and Maks [1985], Afriat [1973], Varian [1982], Varian [1990], Famulari [1995], Färe and Grosskopf [1995], Fleissig and Whitney [2005], Echenique, Lee, and Shum [2011], Apesteguia and Ballester [2015], Castillo and Freer [2016], Dean and Martin [2016], Aguiar and Serrano [2017], Dembo [2017], Heufer and Hjertstrand [2017], Dzielwski [2020], de Clippel and Rozen [2018], Echenique, Imai, Saito, et al. [2018], and Allen and Rehbeck [2019].

³In particular, we use the data sets from Choi, Fisman, Gale, and Kariv [2007], Fisman, Kariv, and Markovits [2007], Choi, Kariv, Müller, and Silverman [2014], Halevy, Persitz, and Zrill [2018], and Feldman and Rehbeck [2020].

⁴For example, on household consumption models [Chiappori, 1988, Cherchye, De Rock, and Vermeulen, 2007, Cherchye and Vermeulen, 2008, Cherchye, De Rock, and Vermeulen, 2009, 2011a], general equilibrium models [Carvajal and Song, 2018, Cherchye, Demuyck, and De Rock, 2011b], counterfactual analysis [Adams, 2020], and tests of weak separability [Cherchye, Demuyck, De Rock, and Hjertstrand, 2015]. For a general review of other computational results and issues in revealed preference, we recommend the interested reader to Smeulders, Crama, and Spieksma [2019].

Outline The remainder of the paper proceeds as follows: Section 2 introduces definitions and notation. Section 3 shows how to check whether data satisfies the generalized axiom of revealed preferences using MILP methods. Section 4 shows how to set up MILPs to compute the Houtman-Maks index, the Average Varian index, and the Minimum Cost Index. Section 5 discusses how to compute the minimal measurement errors on expenditures, prices, and quantities that rationalize the data. Section 6 discusses how the results can be extended to settings with non-linear budgets. Section 7 illustrates our findings by computing the HMI, AVI, and MCI for a multitude of data sets from experiments in the literature. Section 8 contains some final remarks.

2 Notation and definitions

We consider the standard consumer problem with K goods. A decision maker chooses a bundle $q \in \mathbb{R}_+^K$ subject to a linear budget constraint where prices are given by a vector $p \in \mathbb{R}_{++}^K$. Total expenditure is equal to pq .⁵ Consider a finite dataset with T observations denoted by $\mathcal{D} = (q^t, p^t)_{t \leq T}$. Here, q^t is the consumption bundle chosen by a DM at observation t facing the prices p^t .

Definition 1 (Rationalizability). *The dataset $\mathcal{D} = (q^t, p^t)_{t \leq T}$ is rationalizable when there exists a locally non-satiated utility function $u : \mathbb{R}_+^K \rightarrow \mathbb{R}$ such that for all bundles $q \in \mathbb{R}_+^K$ and all $t \leq T$:*

$$p^t q^t \geq p^t q \text{ implies that } u(q^t) \geq u(q).$$

In words, any less expensive bundle at observation t provides less (or equal) utility compared to q^t .

Afriat [1967] and Varian [1982] derived necessary and sufficient conditions for rationalizability. Towards this end, we say the bundle q^t is *revealed preferred* to bundle q^v when $p^t q^t \geq p^t q^v$. We write this as $q^t R q^v$.

Similarly, we say the bundle q^t is *strictly revealed preferred* to q^v when $p^t q^t > p^t q^v$. This is a strict revealed preference since the bundle q^v costs strictly less than the total expenditure at observation t . We write this as $q^t P q^v$. Notice that $P \subseteq R$. We now define the generalized axiom of revealed preference from Varian [1982].

Definition 2 (GARP). *A dataset $\mathcal{D} = (p^t, q^t)_{t \leq T}$ satisfies the Generalized Axiom of Revealed Preference (GARP) when for all sequences of observations $t_1, \dots, t_M \leq T$:*

$$q^{t_1} R q^{t_2} R \dots R q^{t_M} \text{ implies that not } q^{t_M} P q^{t_1}.$$

⁵For vectors $x, y \in \mathbb{R}^K$, we assume that any products are inner products so that $xy = \langle x, y \rangle = \sum_{k=1}^K x_k y_k$.

A well known result in revealed preference theory states that a dataset is rationalizable if and only if GARP is satisfied [Afriat, 1967, Varian, 1982].

Theorem 1 (Afriat's theorem). *A dataset $\mathcal{D} = (p^t, q^t)_{t \leq T}$ is rationalizable if and only if it satisfies GARP.*

The following result is known in the literature, but we provide a proof for completeness.⁶

Theorem 2. *The dataset $\mathcal{D} = (p^t, q^t)_{t \leq T}$ satisfies GARP if and only if for all observations $t \in T$, there exists a number $u_t \in [0, 1]$ such that for all $t, v \leq T$:*

$$p^t q^t \geq p^t q^v \text{ implies } u_t \geq u_v, \quad (\text{GARP-1})$$

$$p^t q^t > p^t q^v \text{ implies } u_t > u_v. \quad (\text{GARP-2})$$

Proof. Let $\mathcal{D}$ satisfy GARP. Let R be the corresponding revealed preference relation and let R^* be its transitive closure. The transitive closure is defined so that $q^t R^* q^v$ if and only if there is a (possibly empty) sequence of observations $t_1, \dots, t_M \leq T$ such that:

$$q^t R q^{t_1} R q^{t_2} \dots R q^{t_M} R q^v.$$

Let $P(R^*)$ be the asymmetric part of R^* . In other words, $q^t P(R^*) q^v$ when $q^t R^* q^v$ and not $q^v R^* q^t$. Notice that if $q^t P q^v$ (i.e. $p^t q^t > p^t q^v$), then it is not the case that $q^v R^* q^t$ as this would imply a violation of GARP. As such, we have that $q^t P q^v$ implies $q^t P(R^*) q^v$ or equivalently $P \subseteq P(R^*)$.

By a variant on Szpilrajn's lemma [Szpilrajn, 1930], we can extend the relation R^* to a transitive, reflexive and complete binary relation $\tilde{R}$ on $(q^t)_{t \leq T}$ that preserves the strict relation, so $R^* \subseteq \tilde{R}$ and $P(R^*) \subseteq P(\tilde{R})$.

As the number of observations T is finite, we can find numbers $u_t \in [0, 1]$ such that:

$$u_t \geq u_v \text{ if and only if } q^t \tilde{R} q^v.$$

We show that these numbers satisfy the conditions in the statement of the theorem. If $p^t q^t \geq p^t q^v$ (i.e. $q^t R q^v$), then $q^t R^* q^v$ which immediately gives $u_t \geq u_v$. Next, if $p^t q^t > p^t q^v$ (i.e. $q^t P q^v$), then we have that $q^t P(\tilde{R}) q^v$ and not $q^v \tilde{R} q^t$. This means that $u^t \geq u^v$ and not $u^v \geq u^t$ which gives $u_t > u_v$.

⁶The formulation in the theorem is very similar to the strong axiom of cost minimization from Varian [1984]. See also Cherchye, Demuynck, De Rock, and Hjertstrand [2015] who use this formulation to analyse the revealed preference conditions for weakly separable preferences.

Next, we show the reverse. Assume that the conditions in the statement of the theorem are satisfied and, towards a contradiction, that GARP is violated. Then, there is a sequence $t_1, \dots, t_M \leq T$ such that

$$q^{t_1} R q^{t_2} R q^{t_3} \dots R q^{t_M} \text{ and } q^{t_M} P q^{t_1}.$$

The conditions in the statement of the theorem then give

$$u_{t_1} \geq u_{t_2} \geq \dots \geq u_{t_M} > u_{t_1},$$

a contradiction. □

Notice that for the conditions in Theorem 2, the restriction $u_t \in [0, 1]$ is arbitrary. Indeed, we could have chosen any arbitrary non-empty interval $u_t \in [a, b]$ (with $a < b$).

3 Testing GARP using via integer programming

Conditions (GARP-1)-(GARP-2) in Theorem 2 give an alternative test for GARP. In particular, we can formalize the “if ... then” into a set of linear restrictions using an integer programming (IP) formulation. Although IP methods are less efficient than other known methods to test for GARP, the new IP formulation will be very useful in terms of formulating various goodness-of-fit measures.

To this end, let $(U_{t,v})_{t,v \leq T}$ be a set of binary variables, i.e. $U_{t,v} \in \{0, 1\}$ for all $t, v \leq T$. The idea is that $U_{t,v} = 1$ if and only if $u_t \geq u_v$ in conditions (GARP-1) and (GARP-2). Let $0 < \varepsilon < \frac{1}{T}$. In order to establish this condition, we impose the following linear restrictions with $u_t \in [0, 1]$ for all $t \leq T$.

$$u_t - u_v \leq -\varepsilon + 2U_{t,v}, \tag{IP-1}$$

$$U_{t,v} - 1 \leq u_t - u_v. \tag{IP-2}$$

Inequality (IP-1) guarantees that $u_t \geq u_v$ implies $U_{t,v} = 1$. To see this, notice that $u_t, u_v \in [0, 1]$ and $u_t \geq u_v$ requires the left hand side of (IP-1) to be greater or equal to zero and less than one. As such, $U_{t,v} = 1$. The factor two on $U_{t,v}$ guarantees that even when $u_t = 1$ and $u_v = 0$ the inequality still holds for $U_{t,v} = 1$. Next, inequality (IP-2) guarantees that $U_{t,v} = 1$ implies $u_t \geq u_v$. Indeed, $U_{t,v} = 1$ imposes that the left hand side of (IP-2) equals zero, which, in turn, imposes that $u_t \geq u_v$.

Next, we need to incorporate the restrictions from (GARP-1) and (GARP-2). For this,

we introduce fixed parameters α and δ . Here $\alpha > \max_{t \leq T} p^t q^t$ and

$$0 < \delta \leq \min \left\{ \min_{t,v} \{p^t q^v\}, \min_{t,v} \{p^t q^v - p^t q^t | p^t q^v > p^t q^t\} \right\}.$$

We use the following two inequalities:

$$p^t q^t - p^t q^v \leq -\delta + \alpha U_{t,v}, \quad (\text{IP-3})$$

$$\alpha(U_{v,t} - 1) \leq p^t q^v - p^t q^t. \quad (\text{IP-4})$$

Here inequality (IP-3) guarantees that if $p^t q^t \geq p^t q^v$, then $U_{t,v} = 1$. Indeed, if $U_{t,v} = 0$, then we would get the contradiction

$$\delta \leq p^t q^v - p^t q^t \leq 0.$$

The fact that $U_{t,v} = 1$ in turn implies that $u_t \geq u_v$ by (IP-2). This yields condition (GARP-1).

Inequality (IP-4) asserts that if $U_{v,t} = 1$ (so $u_v \geq u_t$ and therefore not $u_t > u_v$), then $p^t q^v \geq p^t q^t$ (i.e. not $p^t q^t > p^t q^v$) so we get the contrapositive of condition (GARP-2). The conditions on the parameter α and δ ensure that the inequalities are otherwise non-informative. We summarize the equivalence of GARP to the IP inequalities in the following corollary.

Corollary 1. *The dataset $\mathcal{D} = (p^t, q^t)_{t \leq T}$ satisfies GARP if and only if for all $t, v \leq T$ there are numbers $u_t \in [0, 1]$ and binary variables $U_{t,v} \in \{0, 1\}$ such that the system of linear inequalities (IP-1)-(IP-4) is feasible.*

Proof. That feasibility of (IP-1)-(IP-4) implies consistency with GARP has been shown in the main text. For the reverse, assume that GARP, i.e. (GARP-1) and (GARP-2), is satisfied. Let u_t be the corresponding utility numbers. Without loss of generality, we can transform the numbers u_t such that the distance between any two distinct values u_t and u_v is no less than $\frac{1}{T}$.

Set $U_{t,v} = 1$ if and only if $u_t \geq u_v$. We show that all inequalities (IP-1)-(IP-4) are satisfied.

If $u_t \geq u_v$ then (IP-1) and (IP-2) are satisfied with $U_{t,v} = 1$. If $u_t < u_v$, then $u_t - u_v < -\frac{1}{T} < -\varepsilon$ and $U_{t,v} = 0$, so (IP-1) and (IP-2) are also satisfied. Next, for (IP-3), if $p^t q^t \geq p^t q^v$ then as $u_t \geq u_v$, we have $U_{t,v} = 1$ and condition (IP-3) is satisfied as (using $\delta \leq p^t q^v$ and $\alpha > p^t q^t$)

$$p^t q^t - p^t q^v \leq p^t q^t - \delta < \alpha U_{t,v} - \delta.$$

If $p^t q^t < p^t q^v$ then $U_{t,v} = 0$, and condition (IP-3) holds since

$$p^t q^t - p^t q^v \leq -\delta = -\delta + \alpha U_{t,v}.$$

using that $\delta \leq p^t q^v - p^t q^t$. For (IP-4), if $p^t q^v < p^t q^t$ then $u_t > u_v$ so $U_{v,t} = 0$, which means that (IP-4) is satisfied. If $p^t q^v \geq p^t q^t$ then (IP-4) is satisfied independently of the value of $U_{v,t}$. $\square$

4 Goodness-of-fit measures

Corollary 1 provides a method to check whether GARP is satisfied. In particular, one only needs to verify whether the system of linear inequalities (IP-1)-(IP-4) is feasible or not. This can be done using integer programming solvers. In general, this is a very inefficient method to test for GARP as IP methods are very inefficient compared to alternative available tests for GARP (either via Afriat inequalities [Diewert, 1973] or via Warshall's algorithm [Warshall, 1962, Varian, 1982]).

On the other hand, the advantage of using inequalities (IP-1)-(IP-4) is that they are very versatile. The reason is that the inequalities (IP-1)-(IP-4) are linear in utility numbers, prices, and quantities. This feature can be exploited to compute various goodness-of-fit measures. We subsequently demonstrate this versatility by constructing MILP formulations for the Houtman-Maks index (HMI), the Average Varian Index (AVI), and the Minimum Cost Index (MCI).

4.1 Houtman-Maks index

The Houtman-Maks index gives the relative size of the largest subset of observations that is still consistent with GARP. If the entire dataset is consistent with GARP, then the HMI equals one. If every pair of observations generates a GARP violation, then the largest GARP-consistent subset is one observation so the HMI equals $\frac{1}{T}$. For a finite set A , let us denote by $|A|$ the size of A .

Definition 3. *The Houtman-Maks index of a dataset $\mathcal{D} = (p^t, q^t)_{t \leq T}$ is defined as*

$$\text{HMI}(\mathcal{D}) = \max_{A \subseteq \{1, \dots, T\}} \frac{|A|}{T} \text{ s.t. } (p^t, q^t)_{t \in A} \text{ satisfies GARP.} \quad (\text{HMI})$$

Computing the HMI is an NP-hard problem [Smeulders, Spieksma, Cherchye, and De Rock, 2014], which means that there exists no polynomial time algorithm to compute this index

(unless $P = NP$).

To provide a MILP formulation for the HMI, we start by modifying the GARP characterization from conditions (GARP-1) and (GARP-2) of Theorem 2 in the following way:

$$p^t q^t \geq p^t q^v \text{ and } t \in A \text{ implies } u_t \geq u_v, \quad (\text{HMI-1})$$

$$p^t q^t > p^t q^v \text{ and } t \in A \text{ implies } u_t > u_v. \quad (\text{HMI-2})$$

The following Lemma shows that this condition is equivalent to testing GARP on the subset of observations in A .

Lemma 1. *For the data set $D = (p^t, q^t)_{t \leq T}$, there exist numbers $u^t \in [0, 1]$ ($t \leq T$) such that (HMI-1) and (HMI-2) are satisfied if and only if $(p^t, q^t)_{t \in A}$ satisfies GARP.*

Proof. Assume that $(p^t, q^t)_{t \in A}$ satisfies GARP. Given Theorem 2 we can find for all $t, v \in A$ numbers $u_t \in [0.5, 1]$ such that

$$p^t q^t \geq p^t q^v \text{ and } t, v \in A \text{ implies } u_t \geq u_v,$$

$$p^t q^t > p^t q^v \text{ and } t, v \in A \text{ implies } u_t > u_v.$$

Here we adjusted the bounds on u_t to the interval $[0.5, 1]$ which can be done without loss of generality. For all other $t \in T \setminus A$, set $u_t = 0$.

To show condition (HMI-1), assume that $p^t q^t \geq p^t q^v$ and $t \in A$. If $v \in A$, then immediately $u_t \geq u_v$, so we are done. If $v \notin A$ then $u_t \geq 0.5 > 0 = u_v$ so again condition (HMI-1) is satisfied. If $p^t q^t > p^t q^v$, a similar reasoning holds to show that (HMI-2) holds.

For the reverse, it suffices to restrict conditions (HMI-1) and (HMI-2) to have both $t, v \in A$. The argument of Theorem 2 then shows that conditions (HMI-1) and (HMI-2) imply GARP for the dataset $(p^t, q^t)_{t \in A}$. $\square$

Next, we rephrase (HMI-1) and (HMI-2) as a MILP. Let us introduce a set of additional binary variables $A_t \in \{0, 1\}$. The idea is that $A_t = 1$ if and only if $t \in A$. Conditions (HMI-1) and (HMI-2) can then be formalized, in the presence of inequalities (IP-1) and (IP-2), via the inequalities

$$p^t q^t - p^t q^v \leq -\delta + \alpha [U_{t,v} + (1 - A_t)], \quad (\text{IP-5})$$

$$\alpha [U_{v,t} + A_t - 2] \leq p^t q^v - p^t q^t \quad (\text{IP-6})$$

where α and δ are constants with $\alpha > \max_{t,v \in T} p^t q^t$ and

$$0 < \delta < \min \left\{ \min_{t,v} \{p^t q^v\}, \min_{t,v} \{p^t q^v - p^t q^t | p^t q^v > p^t q^t\} \right\}.$$

The restriction (IP-5) shows that if $p^t q^t \geq p^t q^v$ and $t \in A$ (i.e. $A_t = 1$), then $U_{t,v} = 1$ (i.e. $u_t \geq u_v$). The restriction (IP-6) imposes that if $U_{v,t} = 1$ (i.e. not $u_t > u_v$) and $t \in A$ (i.e. $A_t = 1$), then $p^t q^v \geq p^t q^t$ (i.e. not $p^t q^t > p^t q^v$). Thus, the HMI can be computed using the mixed integer linear programming constraints.

Corollary 2. *The following MILP computes the Houtman-Maks index for the dataset $\mathcal{D} = (q^t, p^t)_{t \leq T}$:*

$$\text{HMI}(\mathcal{D}) = \max_{u_t, A_t, U_{t,v}} \sum_{t \in T} \frac{A_t}{T},$$

subject to inequalities (IP-1), (IP-2), (IP-5), and (IP-6) where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$ and $A_t \in \{0, 1\}$.

4.2 Average Varian Index

A second popular measure of choice consistency is the Average Varian Index [Varian, 1990]. In order to define this index, let us introduce a vector $\mathbf{e} = (e_t)_{t \leq T}$ whose elements take values in the unit interval $[0, 1]$. The idea is that $\mathbf{e}$ gives the degree of relaxation of the revealed preference relation. Towards this end, we define the modified revealed preference relation $q^t R_{\mathbf{e}} q^v$ whenever $e_t p^t q^t \geq p^t q^v$, and the modified strict revealed preference relation $q^t P_{\mathbf{e}} q^v$ when $e_t p^t q^t > p^t q^v$.

We say that a dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ satisfies $\mathbf{e}$ -GARP when for all sequences $t_1, \dots, t_M \leq T$:

$$q^{t_1} R_{\mathbf{e}} q^{t_2} R_{\mathbf{e}} q^{t_3} \dots R_{\mathbf{e}} q^{t_M} \text{ implies that not } q^{t_M} P_{\mathbf{e}} q^{t_1}.$$

If for all $t \in T$ the terms $e_t = 1$, then this gives the usual GARP condition. The lower the values of e_t the less stringent $\mathbf{e}$ -GARP.

Various types of Varian indices can be obtained by maximizing some increasing function of $\mathbf{e}$ subject to $\mathbf{e}$ -GARP. Here, we focus on the average Varian index (AVI) which maximizes the average of the vector $\mathbf{e}$.

Definition 4. *The average Varian index of a dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ is defined as*

$$\text{AVI}(\mathcal{D}) = \sup_{\mathbf{e} \in [0,1]^T} \sum_{t \leq T} \frac{e_t}{T} \text{ subject to } (p^t, q^t)_{t \leq T} \text{ satisfies } \mathbf{e} - \text{GARP}. \quad (\text{AVI})$$

Computing the Varian Index is an NP-hard problem [Smeulders, Spieksma, Cherchye, and De Rock, 2014].⁷ To obtain a feasible method to compute the AVI, we again reformulate it as a MILP. Towards this end, we rephrase the conditions (GARP-1) and (GARP-2) to

$$e_t p^t q^t \geq p^t q^v \text{ implies } u_t \geq u_v, \quad (\text{VI-1})$$

$$e_t p^t q^t > p^t q^v \text{ implies } u_t > u_v. \quad (\text{VI-2})$$

The following lemma shows that **e**-GARP is equivalent to the existence of values $u_t \in [0, 1]$ that satisfy conditions (VI-1) and (VI-2).

Lemma 2. *A dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ satisfies **e**-GARP if and only if there exist numbers $u_t \in [0, 1]$ such that (VI-1) and (VI-2) are satisfied.*

Proof. This proof is similar to Theorem 2. Suppose R_e satisfies **e**-GARP. Take the transitive closure of R_e denoted R_e^* where $P(R_e^*)$ is the asymmetric part of R_e^* . Notice that if $q^t P_e q^v$ (i.e. $e_t p^t q^t > p^t q^v$), then it is not the case that $q^v R_e^* q^t$ as this would imply a violation of **e**-GARP. As such, we have that $q^t P_e q^v$ implies $q^t P(R_e^*) q^v$ or equivalently $P_e \subseteq P(R_e^*)$.

Again, we extend the relation R_e^* to a transitive, reflexive, and complete binary relation on $(q^t)_{t \leq T}$ that preserves the strict relation. Let $\tilde{R}$ be such relation. Since T is finite, there are numbers $u_t, u_v \in [0, 1]$ such that $u_t \geq u_v$ if and only if $q^t \tilde{R} q^v$ where $R_e^* \subseteq \tilde{R}$ and $P(R_e^*) \subseteq P(\tilde{R})$.

We show that these numbers satisfy the conditions in the statement of the lemma. If $e_t p^t q^t \geq p^t q^v$ (i.e. $q^t R_e q^v$), then $q^t R_e^* q^v$, which immediately gives $u_t \geq u_v$ since $\tilde{R}$ extends R_e^* . Next, if $e_t p^t q^t > p^t q^v$, (i.e. $q^t P_e q^v$), then from the reasoning above, we have that $q^t P(R_e^*) q^v$, so also $q^t P(\tilde{R}) q^v$. Recall that $q^t P_e q^v$ implies that it is not $q^v R_e^* q^t$ which gives $u_t > u_v$.

We show the reverse. Suppose **e**-GARP is violated. Thus, there is a sequence $t_1, \dots, t_M$ of observations such that

$$q^{t_1} R_e q^{t_2} R_e q^{t_3} \dots R_e q^{t_M} \text{ and } q^{t_M} P_e q^{t_1}.$$

The conditions in the statement of the theorem give

$$u_{t_1} \geq u_{t_2} \dots \geq u_{t_M} > u_1,$$

a contradiction. □

⁷An exception is when the objective function equals $\min_t e_t$. In this case, the Varian Index coincides with the Afriat Critical Cost Efficiency Index, which can be efficiently computed.

The conditions (VI-1) and (VI-2) can be rephrased as a set of IP inequalities. In the presence of the inequalities (IP-1) and (IP-2), we rephrase the conditions (VI-1) and (VI-2) as

$$e_t p^t q^t - p^t q^v \leq \alpha U_{t,v}, \quad (\text{IP-7})$$

$$\alpha(U_{v,t} - 1) \leq p^t q^v - e_t p^t q^t \quad (\text{IP-8})$$

where $\alpha > \max_{t \leq T} p^t q^t$. Notice that the inequalities (IP-7) and (IP-8) are linear in the unknowns e_t and $U_{t,v}$. The inequalities follow from similar arguments as justifying the inequalities (IP-5) and (IP-6). However, notice that we do not subtract the parameter δ in (IP-7). This results from the fact that the AVI is computed as the supremum instead of the max.

The following result records how to compute the Average Varian Index.

Corollary 3. *The following MILP computes the Average Varian Index for the dataset $\mathcal{D} = (q^t, p^t)_{t \leq T}$:*

$$\text{AVI}(\mathcal{D}) = \max_{u_t, \mathbf{e}, U_{t,v}} \sum_{t \leq T} \frac{e_t}{T},$$

subject to the inequalities (IP-1), (IP-2), (IP-7) and (IP-8) where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$, and $e_t \in [0, 1]$.

Proof. Let $\mathbf{e}$ solve the maximisation problem in the corollary, let $w = \sum_t \frac{e_t}{T}$, and let $v = \text{AVI}(\mathcal{D})$. We need to show that $v = w$. First, As v is a supremum, there must exist a sequence $(\hat{\mathbf{e}}^n, u_t^n)_{t \leq T}$ ($n \in \mathbb{N}$) that satisfy the conditions (VI-1) and (VI-2) and such that:

$$\sum_{t \leq T} \frac{\hat{e}_t^n}{T} \rightarrow v.$$

From Lemma 2, we see that there is a sequence of values $(u_t^n)_{t \leq T, n \in \mathbb{N}}$ between 0 and 1, that together with $\hat{\mathbf{e}}^n$ satisfy (VI-1) and (VI-2). Define for all $n \in \mathbb{N}$ and $t, v \leq T$ the binary variables $U_{t,v}^n \in \{0, 1\}$ such that $U_{t,v}^n = 1$ if and only if $u_t^n \geq u_v^n$. One can check that $(u_t^n, U_{t,v}^n)_{t \leq T}$ satisfy (IP-1), (IP-2), (IP-7), and (IP-8) for all $n \in \mathbb{N}$.

Moving along a convergent subsequence where the binary variables $U_{t,v}^n$ are constant, one can easily verify that these weak inequalities must then also be satisfied in the limit, so:

$$v \leq w.$$

For the reverse, assume, towards a contradiction that $v < w$. Let $(e_t, u_t, U_{t,v})_{t,v \leq T}$ be the solution to the maximisation problem in Corollary 3. Let $\varepsilon > 0$ be a strict positive number

such that:

$$v < \sum_{t \leq T} \frac{e_t(1 - \varepsilon)}{T}.$$

Let us show that $\hat{\mathbf{e}}^t = e_t(1 - \varepsilon)$ together with u_t satisfies (VI-1) and (VI-2).

If $\hat{e}_t p^t q^t - p^t q^v \geq 0$, then $e_t > 0$ and $e_t > \hat{e}_t$, so $e_t p^t q^t - p^t q^v > 0$. From (IP-7) it follows that $U_{t,v} = 1$ and from (IP-1) it follows that $u_t \geq u_v$. Next, for (VI-2) assume that $\hat{e}^t p^t q^t > p^t q^v$. It follows again that $e^t p^t q^t > p^t q^v$, and so by (IP-8) it follows that $U_{v,t} = 0$ and by (IP-1) it follows that $u_t > u_v$. As v is the supremum, we must have that:

$$\sum_{t \leq T} \frac{e_t(1 - \varepsilon)}{T} \leq v,$$

which gives the desired contradiction. $\square$

An popular alternative for the AVI is the index obtained by minimizing the squared distance between $\mathbf{e}$ and the unit vector.

$$\inf_{\mathbf{e} \in [0,1]^T} \sum_{t \leq T} \frac{(1 - e_t)^2}{T} \text{ subject to } (p^t, q^t)_{t \leq T} \text{ satisfies } \mathbf{e} - GARP.$$

Following the same lines as in Corollary 3, one can show that this index can be computed by solving a quadratic objective function:

$$\min_{u_t, \mathbf{e}, U_{t,v}} \sum_{t \leq T} \frac{(1 - e_t)^2}{T},$$

subject to the inequalities (IP-1), (IP-2), (IP-7) and (IP-8) where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$, and $e_t \in [0, 1]$.

4.3 Minimal Cost Index

A final goodness-of-fit measure which demonstrates the use of the MILP formulation is the minimum cost index of Dean and Martin [2016]. The idea behind the minimal cost index is similar in spirit to the Houtman-Maks Index. While the Houtman-Maks Index removes observations to obtain a rationalizable subset of the original dataset, the idea behind the Minimal Cost Index is to remove elements of the revealed preference relation. To do this, the MCI first assigns a cost to each revealed preference relation and then removes those relations that give the largest violations in accordance to this cost.

Towards this end, let B be a subset of the revealed preference relation R so that $q^t B q^v$ implies $q^t R q^v$. The idea is to remove pairs (q^t, q^v) with $q^t B q^v$ from the revealed preference

relation R . We define $(R \setminus B)$ as the relation where $q^t (R \setminus B) q^v$ when $q^t R q^v$ and not $q^t B q^v$. Similarly, define $q^t (P \setminus B) q^v$ when $q^t P q^v$ and not $q^t B q^v$. We say that the dataset satisfies $(R \setminus B)$ -GARP if for all sequences $t_1, \dots, t_M \leq T$:

$$q^{t_1} (R \setminus B) q^{t_2} (R \setminus B) q^{t_3} \dots (R \setminus B) q^{t_M} \text{ implies not } q^{t_M} (P \setminus B) q^{t_1}.$$

Essentially, this says that GARP holds when we omit the comparisons in B .

Definition 5. The minimum cost index of a dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ is defined as:

$$\text{MCI}(\mathcal{D}) = \min_{B \subseteq R} \sum_{t \in T} \frac{p^t(q^t - q^v)}{\sum_{w \in T} p^w q^w} \text{ subject to } (p^t, q^t)_{t \in T} \text{ satisfies } (R \setminus B)\text{-GARP.} \quad (\text{MCI})$$

For the MCI, the cost to remove the relation $q^t R q^v$ is given by $\frac{p^t(q^t - q^v)}{\sum_{w \in T} p^w q^w}$. The goal is to remove only those relations that minimize the total cost.

To formulate this as a MILP, we first formalize the restriction of (MCI) by modifying the conditions (GARP-1) and (GARP-2) to

$$p^t q^t \geq p^t q^v \text{ and } (t, v) \notin B \text{ implies } u_t \geq u_v, \quad (\text{MCI-1})$$

$$p^t q^t > p^t q^v \text{ and } (t, v) \notin B \text{ implies } u_t > u_v. \quad (\text{MCI-2})$$

These restrictions effectively limit the GARP condition to pairs of observations (t, v) that are not in B .

Lemma 3. A dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ satisfies $(R \setminus B)$ -GARP if and only if there exist numbers $u_t \in [0, 1]$ such that (MCI-1) and (MCI-2) are satisfied.

Proof. This proof is similar to Theorem 2. Assume that $(p^t, q^t)_{t \in T}$ satisfies $(R \setminus B)$ -GARP. Again, the transitive closure $(R \setminus B)^*$ can be extended to a transitive, reflexive, and complete binary relation by a variant of Szpilrajn's lemma on $(q^t)_{t \in T}$ that preserves the strict relation. Let $\tilde{R}$ be the extension of $(R \setminus B)$. An important feature is that $q^t (P \setminus B) q^v$ implies that it is not the case that $q^v (R \setminus B)^* q^t$ as this would imply a violation of $(R \setminus B)$ -GARP. This implies that $(P \setminus B) \subseteq P((R \setminus B)^*) \subseteq P(\tilde{R})$ where $P(\cdot)$ gives the asymmetric part of a binary relation. Moreover, we have $(R \setminus B) \subseteq \tilde{R}$ by construction.

Since T is finite, there are numbers $u_t, u_v \in [0, 1]$ such that $u_t \geq u_v$ if and only if $q^t \tilde{R} q^v$. We show that these numbers satisfy the conditions in the statement of the lemma. If $q^t (R \setminus B) q^v$, then $q^t \tilde{R} q^v$, which immediately gives $u_t \geq u_v$. Next, if $q^t (P \setminus B) q^v$, then $q^t P(\tilde{R}) q^v$. Recall that $q^t (P \setminus B) q^v$ implies that it is not $q^v (R \setminus B)^* q^t$ which gives $u_t > u_v$.

We show the reverse. Suppose $(R \setminus B)$ -GARP is violated. Thus, there is a sequence $t_1, \dots, t_M \leq T$ of observations such that

$$q^{t_1} (R \setminus B) q^{t_2} (R \setminus B) q^{t_3} \dots q^{t_{M-1}} (R \setminus B) q^{t_M} \text{ and } q^{t_M} (P \setminus B) q^{t_1}.$$

The conditions in the statement of the theorem give

$$u_{t_1} \geq u_{t_2} \geq \dots \geq u_{t_M} > u_1,$$

a contradiction. □

Again, the conditions (MCI-1) and (MCI-2) can be written using a MILP formulation. To do this, we add binary variables $B_{t,v} \in \{0, 1\}$ with the idea that $B_{t,v} = 1$ if and only if $q^t B q^v$. In the presence of inequalities (IP-1) and (IP-2), we include the inequalities

$$p^t q^t - p^t q^v \leq -\delta + \alpha(U_{t,v} + B_{t,v}), \quad (\text{IP-10})$$

$$\alpha(U_{v,t} - 1 - B_{t,v}) \leq p^t q^v - p^t q^t, \quad (\text{IP-11})$$

$$\beta(B_{t,v} - 1) \leq p^t q^t - p^t q^v \quad (\text{IP-12})$$

where α, β and δ are parameters such that $\alpha > \max_{t \leq T} p^t q^t$, $\beta > \max_{t,v} p^t q^v$ and

$$0 < \delta < \min \left\{ \min_{t,v} p^t q^v, \min_{t,v} \{p^t q^v - p^t q^t \mid p^t q^v > p^t q^t\} \right\}$$

The restriction of (IP-10) requires that if $p^t q^t \geq p^t q^v$ and $B_{t,v} = 0$, then $U_{t,v} = 1$, so $u^t \geq u^v$. Next, restrictions from (IP-11) require that if $U_{v,t} = 1$ (i.e. not $u_t > u_v$) and $B_{t,v} = 0$, then $p^t q^v \geq p^t q^t$ (i.e. not $p^t q^t > p^t q^v$). Finally, the restriction of (IP-12) requires that if $B_{t,v} = 1$, then $p^t q^t \geq p^t q^v$. In other words, B is a subset of the revealed preference relation. Thus, the MCI can be computed using the mixed integer linear programming constraints.

Corollary 4. *The following MILP computes the Minimal Cost index for the dataset $\mathcal{D} = (q^t, p^t)_{t \leq T}$:*

$$\text{MCI}(\mathcal{D}) = \min_{u_t, U_{t,v}, B_{t,v}} \sum_{t,v \in T} \frac{p^t(q^t - q^v)B_{t,v}}{\sum_{w \in T} p^w q^w}$$

subject to (IP-1), (IP-2), (IP-10), (IP-11), and (IP-12) where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$, and $B_{t,v} \in \{0, 1\}$.

We note that Corollary 4 can be used for any type of linear cost assigned to removing a binary relation. Thus, one could use this strategy to compute alternative measures as suggested in Famulari [1995].

5 Measurement Error

We note that the integer-linear programming inequalities used to compute the average Varian index also allow one easily look for minimal measurement error in expenditures, prices or quantities. These methods are intuitive, but might also be useful for statistical tests that incorporate measurement error (see, for example, [Varian \[1985\]](#) or [Aguilar and Kashaev \[forthcoming\]](#)).

Expenditure error First, we incorporate error in the measured expenditure. For example, consider the relaxed revealed preference relation from non-negative numbers $\mathbf{r} = (r_t)_{t \leq T}$ such that $q^t R_{\mathbf{r}} q^v$ whenever $p^t q^t - r_t \geq p^t q^v$. Similarly, let $q^t P_{\mathbf{r}} q^v$ when $p^t q^t - r_t > p^t q^v$. The idea is to relax the revealed preference condition by introducing expenditure error. For example, a shopper might have coupons for 20% off a purchase which are unobserved.

We define $\mathbf{r}$ -GARP as for all sequences $t_1, \dots, t_M \leq T$ that

$$q^{t_1} R_{\mathbf{r}} q^{t_2} \dots R_{\mathbf{r}} q^{t_M} \text{ implies not } q^{t_M} P_{\mathbf{r}} q^{t_1}.$$

We define the Average Expenditure Error (AEE) to be the smallest average value of $\mathbf{r}$ such that $(p^t, q^t)_{t \leq T}$ satisfies $\mathbf{r}$ -GARP.

Definition 6. *The Average Expenditure Error of a dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ is defined as*

$$\text{AEE}(\mathcal{D}) = \inf_{\mathbf{r}} \sum_{t \leq T} \frac{r_t}{T} \text{ subject to } (p^t, q^t)_{t \in T} \text{ satisfies } \mathbf{r}\text{-GARP}. \quad (\text{AEE})$$

Following the arguments from Section 4.2, we can generate the integer programming inequalities

$$p^t q^t - p^t q^v - r_t \leq \alpha U_{t,v}, \quad (\text{IP-13})$$

$$\alpha(U_{v,t} - 1) \leq p^t q^v - p^t q^t + r_t. \quad (\text{IP-14})$$

One can then find the average expenditure error (AEE) for a dataset from the following corollary.

Corollary 5. *We have that*

$$\text{AEE}(\mathcal{D}) = \min_{\mathbf{r}, u_t, U_{t,v}} \sum_{t \in T} \frac{r_t}{T}$$

subject to (IP-1), (IP-2), (IP-13), and (IP-14) where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$, and $r_t \geq 0$.

Price error Just as there may be errors measuring expenditure, there may also be errors measuring prices. For example, a researcher may only have access to a suggested retail price, but not the price an individual pays. Similar to the expenditure errors, we can introduce errors on prices. Towards this end, we first normalize prices such that for every observation $t \leq T$ we have $\sum_{k=1}^K p_k^t = 1$, where we denote by p_k^t the price of good k at observation t . Next, we introduce unknown normalized prices $(\tilde{p}^t)_{t \leq T}$ where each $\tilde{p}^t \in \mathbb{R}_{++}^K$ and $\sum_{k=1}^K \tilde{p}_k^t = 1$.

The aim is to find such prices $(\tilde{p}^t)_{t \leq T}$ that are as close as possible to p^t and such that the dataset $(\tilde{p}^t, q^t)_{t \leq T}$ satisfies GARP. To measure the distance between prices, we use the L_1 -norm:

$$\|p^t - \tilde{p}^t\| = \sum_{k=1}^K |p_k^t - \tilde{p}_k^t|.$$

We then define the Average Price Error (APE) in the following way:

Definition 7. *The Average Price Error of a dataset $\mathcal{D} = (p^t, q^t)_{t \in T}$ is defined as*

$$\text{APE}(\mathcal{D}) = \frac{1}{T} \inf_{\tilde{\mathbf{p}}} \sum_{t \leq T} \|p^t - \tilde{p}^t\| \text{ subject to } (\tilde{p}^t, q^t)_{t \in T} \text{ satisfies GARP.} \quad (\text{APE})$$

Following the arguments in Section 4.2, we can generate the integer programming inequalities

$$\tilde{p}^t q^t - \tilde{p}^t q^v \leq \alpha U_{t,v}, \quad (\text{IP-15})$$

$$\alpha(U_{v,t} - 1) \leq \tilde{p}^t q^v - \tilde{p}^t q^t \quad (\text{IP-16})$$

where $\alpha > \max_{\substack{t \leq T \\ k \leq K}} q_k^t$.

The L_1 -norm can be computed by linear programming by introducing variables Δ_j^t and adding the constraints:

$$\tilde{p}_j^t - p_j^t \leq \Delta_j^t, \quad (\text{IP-17})$$

$$p_j^t - \tilde{p}_j^t \leq \Delta_j^t. \quad (\text{IP-18})$$

$$\sum_k \tilde{p}_k^t = 1 \quad (\text{IP-19})$$

The conditions (IP-17) and (IP-18) guarantee that $|p_j^t - \tilde{p}_j^t| < \Delta_j^t$. Condition (IP-19) requires that prices add up to one.

We record the following result on the average pricing error (APE) below.

Corollary 6. *We have that*

$$\text{APE}(\mathcal{D}) = \min_{\tilde{\mathbf{p}}, \Delta_k^t, u_t, U_{t,v}} \frac{1}{T} \sum_{t \in T} \frac{\sum_{k=1}^K \Delta_k^t}{\|p^t\|},$$

subject to (IP-1), (IP-2), (IP-15), (IP-16), (IP-17) and (IP-18), where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$, $\tilde{p}^t \in \mathbb{R}_{++}^K$ and $\Delta_k^t \in \mathbb{R}$.

Instead of measurement error on prices, we can conduct a similar setting for measurement error on quantities. For this, one needs to introduce variables $\tilde{q}^t \in \mathbb{R}_+^K$ and define GARP consistency conditional on these bundles. The Average Quantity Error (AQE) would then consist of minimizing the mean of the errors $\|q^t - \tilde{q}^t\|$ conditional on $(p^t, \tilde{q}^t)_{t \leq T}$ satisfying GARP.

6 General budgets

This section considers measures of choice consistency with general budget sets in the spirit of Forges and Minelli [2009]. We allow the budget sets to have fewer conditions than Forges and Minelli [2009], but we also place stronger assumptions on the observed data so that the results of Fostel et al. [2004] still apply. These results apply when budgets do not have a common direction or the boundary of budget sets is non-linear. Here, we focus on computation of the Houtman-Maks index since adjusting the methods above for the AVI and MCI with general budget constraints is less straightforward as both definitions rely crucially on the fact that budgets are linear.

Consider an individual choosing among bundles in $\mathbb{R}_+^K$. Assume that the budget set is defined by a function $g^t : \mathbb{R}_+^K \rightarrow \mathbb{R}_+$. A bundle $q \in \mathbb{R}_+^K$ is feasible for the budget g^t if and only if $g^t(q) \leq 0$. In addition, assume for all observations $t \leq T$ that $g^t(q^t) = 0$. A dataset takes the form $\mathcal{D}_G = (g^t, q^t)_{t \leq T}$.

These budget sets cover the standard consumer problem with linear budgets. To see this, note that for prices p^t , the linear budget can be given as $g^t(q) = p^t q - p^t q^t$. Here, $g^t(q) \leq 0$ when $p^t q \leq p^t q^t$ and $g^t(q) < 0$ when $p^t q < p^t q^t$. We record the definition of rationalisability below.

Definition 8. *A dataset $\mathcal{D}_G$ is strongly rationalisable when there exists a utility function $u : \mathbb{R}_+^K \rightarrow \mathbb{R}$ when for all $t \leq T$ and all $q \in \mathbb{R}_+^K$*

$$g^t(q) \leq 0 \text{ implies } u(q^t) \geq u(q), \quad (\text{SR-1})$$

$$g^t(q) < 0 \text{ implies } u(q^t) > u(q). \quad (\text{SR-2})$$

This definition is similar to the condition in Theorem 2. Relative to the standard rationalizability definition, Definition 8 makes the assumption that observed choices are on the boundary of the budget set and maximize some preference relation.

Using these assumptions, we define a revealed preference relation where $q^t R_g q^v$ whenever $g^t(q^v) \leq 0$. Similarly, we define the strict revealed preference where $q^t P_g q^v$ when $g^t(q^v) < 0$. We say the dataset $\mathcal{D}_g$ satisfies g -GARP when for all sequences $t_1, \dots, t_M \leq T$ that

$$q^{t_1} R_g q^{t_2} R_g q^{t_3} \dots R_g q^{t_M} \text{ implies not } q^{t_M} P_g q^{t_1}.$$

Following the arguments from Theorem 2 applied to the revealed preference relation R_g and P_g , the conditions of a strong rationalization become

$$g^t(q^v) \leq 0 \text{ implies } u_t \geq u_v, \quad (\text{SR-1})$$

$$g^t(q^v) < 0 \text{ implies } u_t > u_v. \quad (\text{SR-2})$$

These conditions will form the basis of the mixed integer linear program.⁸

Using these conditions, we can study measures of choice consistency for more general budget sets. We demonstrate conditions that can check g -GARP. First, we can still use the inequalities (IP-1) and (IP-2) to restrict the u_t terms. Next, we can find inequalities similar to (IP-3) and (IP-4) to implement the conditions (SR-1) and (SR-2). In particular, the inequalities

$$-g^t(q^v) \leq -\delta + \gamma U_{t,v}, \quad (\text{IP-20})$$

$$\gamma(U_{v,t} - 1) \leq g^t(q^v). \quad (\text{IP-21})$$

where $\gamma > \max_{t,v} |g^t(q^v)| + \min\{1, \min_{t,v} \{g^t(q^v) \mid g^t(q^v) > 0\}\}$ and $0 < \delta < \min\{1, \min_{t,v} \{g^t(q^v) \mid g^t(q^v) > 0\}\}$. Restriction (IP-20) guarantees that if $g^t(q^v) \leq 0$, then $U_{t,v} = 1$. This in turn implies $u_t \geq u_v$ by (IP-1) which gives (SR-1). The restriction of (IP-21) asserts that if $u_v \geq u_t$ (i.e. not $u_t > u_v$) so $U_{v,t} = 1$, then $g^t(q^v) \geq 0$ (i.e. not $g^t(q^v) < 0$). This is the contrapositive of (SR-2).

With the information from (IP-20) and (IP-21) we can formulate a MILP to compute

⁸To see how these could be altered for a type of Varian Index, consider $\mathbf{e} \in \mathbb{R}_+^T$ and the revealed preference relation given by $g^t(q^v) + e_t \leq 0$. To see how this could be altered for a type of cost index, consider removing pairs (t, v) of the revealed preference relation by introducing a $B_{t,v}$ binary variable as before.

the HMI. It is easy to see that the analogues to (IP-5) and (IP-6) are given by

$$-g^t(q^v) \leq -\delta + \gamma(U_{t,v} + 1 - A_t), \quad (\text{IP-22})$$

$$\gamma(U_{v,t} + A_t - 2) \leq g^t(q^v). \quad (\text{IP-23})$$

where $A_t \in \{0, 1\}$ is a binary variable that tracks whether observation t is in the largest strong rationalizable subset. We record the method to compute the HMI for general budget sets below.

Corollary 7. *The following MILP computes the Houtman-Maks index for the dataset $\mathcal{D}_G$:*

$$\text{HMI}(\mathcal{D}_g) = \max_{u_t, A_t, U_{t,v}} \sum_{t \in T} \frac{A_t}{T},$$

subject to inequalities (IP-1), (IP-2), (IP-20), and (IP-21) where $u_t \in [0, 1]$, $U_{t,v} \in \{0, 1\}$ and $A_t \in \{0, 1\}$.

7 Illustrative Application

We demonstrate the usefulness of our MILP formulations by computing various measures of choice consistency. The data we use is from Choi, Fisman, Gale, and Kariv [2007] (CFGK 2007), Fisman, Kariv, and Markovits [2007] (FKM 2007), Choi, Kariv, Müller, and Silverman [2014] (CKMS 2014), Halevy, Persitz, and Zrill [2018] (HPZ 2018), and Feldman and Rehbeck [2020] (FR 2020). The datasets (CFGK 2007), (CKMS 2014) and (HPZ 2018) examine demand for risky assets, (FKM 2007) studies dictator games, (FR 2020) studies risk preference in the probability simplex. All of the studies except (FR 2020) have linear budgets from the standard consumer problem. We describe relevant details of these dataset in Table 1.

We set up the MILP using Matlab and solve using Gurobi Optimization [2021]. GurobiTM is a commercial package that can solve linear, integer and mixed integer linear programming problems. It is free for academic use. We present statistics on the running times in Table 2. The program was run on a desktop computer with an Intel(R) Core(TM) i7-7700 CPU 3.60GHz processor with 16.0 GB RAM. The main take away from this illustration is that nearly all measures can be computed in less than one second.

Paper	Treatment	T	Number of Subjects
CFGK 2007	Symmetric	50	47
	Asymmetric I	50	17
	Asymmetric II	50	29
FKM 2007	Two players	50	76
	Three players	50	65
CKMS 2014		25	1182
HPZ 2018		22	207
FR 2020		79	144

Table 1: Details on experimental datasets

Study	Measure	Mean	Median	Max	Min
CFGK 2007 Sym	HMI	0.0981	0.0997	0.1326	0.0746
	AVI	0.2863	0.2210	0.8211	0.1187
	MCI	0.1059	0.1019	0.1376	0.0938
CFGK 2007 Asym I	HMI	0.0941	0.0978	0.1127	0.0748
	AVI	0.2846	0.2331	0.5944	0.1228
	MCI	0.1025	0.0987	0.1336	0.0937
CFGK 2007 Asym II	HMI	0.0975	0.0997	0.1197	0.0761
	AVI	0.3056	0.2503	0.7151	0.1366
	MCI	0.1008	0.0997	0.1208	0.0924
FKM 2007 Two	HMI	0.0974	0.0987	0.1546	0.0738
	AVI	0.2935	0.2005	1.2101	0.1177
	MCI	0.1031	0.1017	0.1546	0.0923
FKM 2007 Three	HMI	0.0952	0.0943	0.1616	0.0728
	AVI	0.2766	0.2064	2.1083	0.1157
	MCI	0.1067	0.1023	0.2131	0.0907
CKMS 2014	HMI	0.0542	0.0539	0.1137	0.0349
	AVI	0.0950	0.0883	0.2882	0.0598
	MCI	0.0581	0.0558	0.1267	0.0459
HPZ 2018	HMI	0.0464	0.0478	0.0878	0.0339
	AVI	0.0677	0.0628	0.1346	0.0538
	MCI	0.0534	0.0508	0.1077	0.0449
FR 2020	HMI	0.2107	0.1962	0.7198	0.1744

Table 2: Details on computation time (in seconds) for various studies and measures

8 Conclusion

This paper demonstrates how to reformulate the computation of many goodness-of-fit revealed preference measures as a MILP problem. This allows us to compute the HMI, AVI, and MCI using standard integer programming methods. Moreover, we discussed how these methods can be used to compute the minimal measurement error on expenditures and prices and how it can be applied in settings with general (nonlinear) budget sets. Through an illustrative application, we showed that, although many of these problems are NP-hard, they can be quickly computed for datasets that are regularly collected in experiments.

References

- Abi Adams. Mutually consistent revealed preference demand predictions. *American Economic Journal: Microeconomics*, 12:42–74, 2020.
- Sidney N Afriat. The construction of utility functions from expenditure data. *International Economic Review*, 8:67–77, 1967.
- Sydney N Afriat. On a system of inequalities in demand analysis: an extension of the classical method. *International Economic Review*, 14:460–472, 1973.
- Victor H Aguiar and Nail Kashaev. Stochastic revealed preferences with measurement error: Testing for exponential discounting in survey data. *Review of Economic Studies*, forthcoming.
- Victor H Aguiar and Roberto Serrano. Slutsky matrix norms: The size, classification, and comparative statics of bounded rationality. *Journal of Economic Theory*, 172:163–201, 2017.
- Roy Allen and John Rehbeck. Assessing misspecification and aggregation for structured preferences. 2019.
- Jose Apesteguia and Miguel A Ballester. A measure of rationality and welfare. *Journal of Political Economy*, 123:1278–1310, 2015.
- Raymond C Battalio, John H Kagel, Robin C Winkler, Edwin B Fisher Jr, Robert L Basmann, and Leonard Krasner. A test of consumer demand theory using observations of individual consumer purchases. *Economic Inquiry*, 11:411–428, 1973.

- Andrés Carvajal and Xinxi Song. Testing pareto efficiency and competitive equilibrium in economies with public goods. *Journal of Mathematical Economics*, 75:19–30, 2018.
- Marco Castillo and Mikhail Freer. A revealed preference test of quasi-linear preferences. Technical report, Working Paper, 2016.
- Laurens Cherchye and Frederic Vermeulen. Nonparametric analysis of household labor supply: goodness of fit and power of the unitary and the collective model. *The Review of Economics and Statistics*, 90:267–274, 2008.
- Laurens Cherchye, Bram De Rock, and Frederic Vermeulen. The collective model of household consumption: a nonparametric characterization. *Econometrica*, 75:553–574, 2007.
- Laurens Cherchye, Bram De Rock, and Frederic Vermeulen. Opening the black box of intrahousehold decision making: Theory and nonparametric empirical tests of general collective consumption models. *Journal of Political Economy*, 117:1074–1104, 2009.
- Laurens Cherchye, Bram De Rock, and Frederic Vermeulen. The revealed preference approach to collective consumption behaviour: Testing and sharing rule recovery. *The Review of Economic Studies*, 78:176–198, 2011a.
- Laurens Cherchye, Thomas Demuynck, and Bram De Rock. Testable implications of general equilibrium models: an integer programming approach. *Journal of Mathematical Economics*, 47:564–575, 2011b.
- Laurens Cherchye, Thomas Demuynck, Bram De Rock, and Per Hjertstrand. Revealed preference tests for weak separability: an integer programming approach. *Journal of Econometrics*, 186:129–141, 2015.
- Pierre-André Chiappori. Rational household labor supply. *Econometrica*, 56:63–90, 1988.
- Syngjoo Choi, Raymond Fisman, Douglas Gale, and Shachar Kariv. Consistency and heterogeneity of individual behavior under uncertainty. *American Economic Review*, 97:1921–1938, 2007.
- Syngjoo Choi, Shachar Kariv, Wieland Müller, and Dan Silverman. Who is (more) rational? *American Economic Review*, 104:1518–50, 2014.
- Geoffroy de Clippel and Kareen Rozen. Consumer theory with misperceived tastes. Technical report, 2018.

- Mark Dean and Daniel Martin. Measuring rationality with the minimum cost of revealed preference violations. *Review of Economics and Statistics*, 98:524–534, 2016.
- Aluma Dembo. Set estimation of an unknown error rate using random revealed preferences. 2017. Working Paper.
- Walter E Diewert. Afriat and revealed preference theory. *Review of Economic Studies*, 40: 419–425, 1973.
- Paweł Dziwulski. Just-noticeable difference as a behavioural foundation of the critical cost-efficiency index. *Journal of Economic Theory*, 188:105071, 2020. Working Paper.
- Federico Echenique, Sangmok Lee, and Matthew Shum. The money pump as a measure of revealed preference violations. *Journal of Political Economy*, 119(6):1201–1223, 2011.
- Federico Echenique, Taisuke Imai, Kota Saito, et al. Approximate expected utility rationalization. Technical report, CRC TRR 190 Rationality and Competition, 2018.
- Melissa Famulari. A household-based, nonparametric test of demand theory. *Review of Economics and statistics*, 77(2):372–382, 1995.
- Rolf Färe and Shawna Grosskopf. Nonparametric tests of regularity, farrell efficiency, and goodness-of-fit. *Journal of Econometrics*, 2(69):415–425, 1995.
- Paul Feldman and John Rehbeck. Revealing a preference for mixtures: An experimental study of risk. *Available at SSRN 3543826*, 2020.
- Raymond Fisman, Shachar Kariv, and Daniel Markovits. Individual preferences for giving. *American Economic Review*, 97(5):1858–1876, 2007.
- Adrian R Fleissig and Gerald A Whitney. Testing for the significance of violations of afriat’s inequalities. *Journal of Business & Economic Statistics*, 23(3):355–362, 2005.
- Francoise Forges and Enrico Minelli. Afriat’s theorem for general budget sets. *Journal of Economic Theory*, 144(1):135–145, 2009.
- Ana Fostel, Herbert E Scarf, and Michael J Todd. Two new proofs of afriat’s theorem. *Economic Theory*, 24(1):211–219, 2004.
- LLC Gurobi Optimization. Gurobi optimizer reference manual, 2021. URL <http://www.gurobi.com>.

- Yoram Halevy, Dotan Persitz, and Lanny Zrill. Parametric recoverability of preferences. *Journal of Political Economy*, 126(4):1558–1593, 2018.
- Jan Heufer and Per Hjertstrand. Consistent subsets: Computationally feasible methods to compute the houtman–maks-index. *Economics Letters*, 128:87–89, 2015.
- Jan Heufer and Per Hjertstrand. Homothetic efficiency: Theory and applications. *Journal of Business & Economic Statistics*, pages 1–13, 2017.
- Hendrik S Houthakker. Revealed preference and the utility function. *Economica*, 17(66):159–174, 1950.
- Martijn Houtman and J Maks. Determining all maximal data subsets consistent with revealed preference. *Kwantitatieve methoden*, 19:89–104, 1985.
- Anthony YC Koo. An empirical test of revealed preference theory. *Econometrica*, 31:646–664, 1963.
- Marcel K Richter. Revealed preference theory. *Econometrica*, 34:635–645, 1966.
- Paul A Samuelson. A note on the pure theory of consumer’s behaviour. *Economica*, 5:61–71, 1938.
- Bart Smeulders, Frits CR Spieksma, Laurens Cherchye, and Bram De Rock. Goodness-of-fit measures for revealed preference tests: Complexity results and algorithms. *ACM Transactions on Economics and Computation (TEAC)*, 2(1):1–16, 2014.
- Bart Smeulders, Yves Crama, and Frits CR Spieksma. Revealed preference theory: An algorithmic outlook. *European Journal of Operational Research*, 272:803–815, 2019.
- Edward Szpilrajn. Sur l’extension de l’ordre partiel. *Fundamentae Mathematicae*, 16:386–389, 1930.
- Hal R Varian. The nonparametric approach to demand analysis. *Econometrica*, 50:945–973, 1982.
- Hal R Varian. The nonparametric approach to production analysis. *Econometrica*, 52:579–597, 1984.
- Hal R Varian. Goodness-of-fit in optimizing models. *Journal of Econometrics*, 46:125–140, 1990.

Hall Varian. Non-parametric analysis of optimizing behavior with measurement error. *Journal of Econometrics*, 30:445–458, 1985.

Stephen Warshall. A theorem of boolean matrices. *Journal of the American Association of Computing Machinery*, 9:11–12, 1962.